

SIMON JUPP

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EDUCATION

Yale University

New Haven, CT

B.S. in Computer Science: GPA 3.8/4.0, Nominated for distinction in major

May 2025

Coursework: Data Structures • Algorithms • Systems Programming & Computer Organization • Object Oriented Programming • Full Stack Web Dev • Computational Intelligence for Games • Machine Learning • Building Game Engines • Real-Time 3D Graphics

SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, SQL, R, Assembly, D
- **Frameworks/Tools:** OpenGL, TensorFlow, Keras, PyTorch, Django, React, NumPy, Git, CUDA, Matplotlib
- **Web Development:** HTML, CSS, Django, React

WORK EXPERIENCE

Enrich Biosystems Inc.

Branford, CT

Software Engineer Intern

June 2024 - Aug. 2024

- Developed a machine learning pipeline to streamline the integration of new training data, incorporating a custom-built GUI to enhance usability and support scalable workflows in a dynamic startup environment
- Created and incorporated computer vision similarity algorithms to highlight specialized cancer and T-cell interactions.

Enrich Biosystems Inc.

Branford, CT

Software Engineer Intern

May 2023 - Dec 2023

- Developed and implemented deep learning image segmentation (ResNet, UNet) models using PyTorch and TensorFlow to enhance the accuracy of cell identification and classification algorithms.
- Conducted data preprocessing and cleaning to prepare large datasets for model training, including data augmentation and normalization techniques.
- Implemented an automated version control pipeline for Sphinx documentation website deployment upon new code releases.

PROJECTS

3D Level-of-Detail Terrain Renderer

- Designed and implemented a real-time terrain rendering engine in **OpenGL** using level-of-detail (LOD) algorithms to dynamically adjust mesh complexity based on camera distance, optimizing performance without sacrificing visual fidelity.
- Leveraged tessellation shaders and Geomipmapping as methods to achieve dynamic LOD and performed benchmark analysis on CPU/GPU load.

Water Simulation

- Engineered a real-time water simulation in **OpenGL** and **D**, using a custom rendering engine.
- Developed advanced graphics features including reflection/refraction, normal mapping, Fresnel effects, and procedural wave generation using custom GLSL shaders to create physically-based water rendering.

Yale Bulldog Buddies

- Created a full-stack app allowing students to find study partners based on a variety of user preferences.
- Developed the backend using **Django** and designed models for profile information.
- Created a **REST API** to allow calls from the frontend to fetch and put data based on user input.

Rhythm Game Engine

- Developed a custom game engine for a rhythm-based game using **D** and **SDL2**, implementing core game design patterns (e.g., observer, state, component) to manage game flow, input handling, and scene transitions.
- Engine includes real-time hit detection and scoring mechanics, allowing users to create, test, and refine custom levels using a modular, user-friendly level editor.

Task Manager

- Built a full-stack application to allow users to plan events and practice time management.
- Used **JSON databases** to simulate a backend and maintain the user's tasks upon restarting the application.
- Used **Javascript**, **React**, **CSS** on the frontend for a dynamic representation of their tasks.

LEADERSHIP & ACTIVITIES

Yale Varsity Track & Field

New Haven, CT

- Train 20 hours per week throughout the academic year while achieving top results in the Ivy League Conference.

Yale Student Athlete Advisory Committee

New Haven, CT

- Collaborated with university administrators to provide actionable insights on the student-athlete experience, influencing policy changes affecting athletic programs.
- Contributed input on new regulations to improve student-athlete welfare, ensuring alignment with NCAA and Ivy League guidelines.